Kaustubh Sakhalkar

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Research Interests

I have a profound interest in the areas of Computer Vision, Statistical Machine Learning and Deep Learning. To be more specific I like to solve problems related to video and scene understanding.

Education

Vellore Institute Technology (Vellore Campus) (2014-2018)
Bachelor of Technology (Computer Science with specilization in Bioinformatics)
Chancellors Special achiever's award for exemplary performance at national / international events and meritorious academic performance. Awarded to students for all round excellence.

Higher Secondary Certificate, School (Maharashtra State Board) (2014)
Secondary School Certificate, School (Maharashtra State Board) (2012)

Work Experience

DreamVu Inc. — Applied Computer Vision Engineer (June 2018 - Present)

Internships

- INRIA Sophia-Antipolis — Computer Vision Intern, STARS team (Jan - May 2018)
  Advised by: Dr Francois Bremond
- Indian Statistical Institute (ISI) Kolkata — Computer Vision Intern (May - July 2017)
  Advised by: Dr Ujjwal Bhattacharya — CVPR Unit
- Gesture Research International Pvt. Ltd (December 16 - January 17)

Publications

- Combining Appearance, Motion & Pose for Human Activity Recognition from RGB-D videos
  Srijan Das, Kaustubh Sakhalkar, Michal Koperski, Francois Bremond, Monique Thonnat.
  To Appear (2019)
- Spatio-Temporal Grids for Daily Living Action Recognition
  Srijan Das, Kaustubh Sakhalkar, Michal Koperski, Francois Bremond
  To Appear (2018)

Technical Skills

Languages: C++, Python — Tools: MATLAB, SQL — Libraries: OpenCV, Tensorflow, Pytorch, Keras, Numpy, Scikit-Learn, Scikit-Image, Pandas — Other: LATEX, bash, Linux, Robotic Operating System (familiar), Raspberry Pi.

Relevant Courses

Object Oriented Programming, Computer Architecture And Organization, Data Structures And Algorithms, Operating Systems, Soft Computing, Agent Based Intelligent Systems, Database Concepts, Image And Vision Computing, Probability And Statistics, Software Engineering, Data Warehousing, Linear Algebra, Automata Theory, Networking.

Undergraduate Projects

- Improving optical flow representations for 3D convolutions (INRIA)
  Proposed a separable convolutional block instead of 3D convolutions for video classification by separating the spatial and temporal convolutions [(2+1)D]. Represented the optical flow as magnitude and orientation in HSV colour space to represent flow (temporal) information. Showed that it improved results for I3D network and improved upon the state of the art. Also conducted abalation studies with frame selection in videos for video representations.

- Featuring the features for action recognition (INRIA)
  Studying the contribution of using multiple video cues for representing appearance, short-term and pose based motion to reach a high action recognition rate. Proposed a two level fusion strategy to combine the features from different cues to address False positives and negatives of similar actions which are common in action recognition by conditional binary classifier to disambiguate them ensuring a sequential refinement of action labels. We use ResNet152 features for appearance, Improved dense trajectories for short term motion and skeleton with LSTM for long term motion. We validate our approach on four public datasets, CAD-60, CAD-120, MSRDailyActivity3D and NTU-RGB+D improving the state-of-the-art results on them.

- Spatio-temporal Action grids (INRIA)
  Addressing the challenge of spatio-temporal relationship in videos for action recognition. Introducing grids in the spatial domain as well as temporal domain for spatio-temporal modeling of features To represent temporal information, we divide the action sequence into temporal segments termed as temporal grids. Cross subject variance is targetted by applying clustering on the appearance features and velocity features to partition the temporal segments. This ensures consistency of the temporal segments for modeling the actions. Results were competitive with the state of the art for four datasets.

- Document Image Binarization (ISI Kolkata)
  Proposed a novel method for degraded document image binarization using a convolutional autoencoder with skip connections and ensembling of three global and local thresholding methods. Learned weighted voting for the decisions from three pipelines to obtain a binary decison for each pixel. Patch to patch binarization technique for document page. Achieved an average F-measure of 90.18 on DIBCO datasets within 1% of the state of the art but on a wider range of document problems. Involved in independent research at the handwriting recognition lab. To the best of my knowledge this was the first time deep learning was used for document binarization.

- European Rover Challenge [SEDS-VIT] [ERC]
  Developed a monocular camera visual odometry system for the rover usinng LK optical flow and FAST algorithm also developed the control interface using Qt GUI tools. Our team was finally placed 20th globally and 2nd in Asia.

- Human activity recognition using 3D convolutions (VIT)
  Implemented a C3D like 3D convolutional neural network in keras and trained it on KTH and weizmann datasets. Compared it with hand crafted techniques like dense trajectories, HOG, STIP. Achieved State of the art results on the two datasets.

- Gesture Controlled Interface for Electronic Appliances (Gesture Research International)
  Developed an API for hand gesture recognition and tracking in real time for microcontrollers. The API was subsequently used by the company in a product controlling electrical appliances.

- Gene study of Gestational Diabetes Mellitus
  Studied and found the most probable genes that cause GDM using the microarray data of the disease using Benjamini Hochberg method. Find the regulation of the genes and their interacting partners. 21 genes were recognized to influence diabetes.

- Real time upper body pose estimation using MS Kinect
  Estimated the pose of the human upper body by applying distance transform to the depth image and using anthropometric ratios of the human body.

- Facial Emotion recognition
  Evaluated the performance of 20 different classification techniques for emotion recognition. Including CNN, SVM and KNN. Feature descriptors used were facial landmarks, Gabor filters, HOG, LBP and LDP.

- Adaptive Skin Segmentation using Histogram back projection
  The project aimed at adaptive skin colour segmentation by detecting faces. The face was recognized with a viola jones classifier and the region of interest was used in histogram back projection to create a mask of skin pixels in the frame.

Miscellaneous Projects

1. Pedestrian detection using histogram of oriented gradients
2. Restaurant management database system using MySql as backend and PyQt4 GUI design
3. A UNIX shell written in C language with history features.
4. File encryption and Decryption using RSA algorithm

Achievements and Other extra-curricular details

- Selected for vision understanding and machine intelligence summer school at INESCTEC Portugal. (2017)(did not attend)
- Second position in WeHack hackathon at VIT University
- Ingenius National hackathon finalist (top 10 out of 300 teams)
- General Electric healthcare hackathon finalist
- Computer Science Co-Head SEDS-VIT student chapter.
- Student Member of Creation Labs.
- Hostel Review committee member. (2016-2017)